SPCC Practical Exam Question Bank

JAVA / C / C++ / Python Programs

1. Program to implement Lexical Analyzer

2. Program to implement Multi-Pass Macroprocessor for the given Assembly Language (Pass1).
Generation of different tables is expected from given sample program

3. Program to remove Left Recursion from the given grammar
Grammar will be given, modified grammar without left recursion is expected

4. Program to perform Left factoring of the given grammar
Grammar will be given, modified grammar with left factoring is expected

5. Program to implement the FIRST set for the given grammar
Grammar will be given, output of FIRST function is expected

6. Program to implement the FOLLOW set for the given grammar
Grammar and FIRST of grammar will be given, output of FOLLOW function is expected

7. Program to implement LL(1) parser
FIRST and FOLLOW will be given, Parsing table is expected

8. Program to generate Three Address Code (TAC)
Input expression will be given, corresponding TAC is expected

9. Program to develop an Optimized Code.
Sample Code will be given, optimized code is expected

10. **Write a program to construct a Symbol Table from the given source code.**
The symbol table should store:

- **Variable name**
- **Type (int, float, etc.)**
- **Size**

Sample Code will be given, symbol table is expected

LEX Programs

1. Write a LEX Program to check whether input character is vowel or consonants.
2. Write a LEX Program to check whether the string is a word or a number.
3. Write a LEX Program to count the number of lines, words and characters in a text.
4. Write a LEX Program to implement Lexical Analyzer.
5. Write a LEX program to check whether a character is a **digit or alphabet**.
6. Write a LEX program to count the number of **vowels in a string**.
7. Write a LEX program to convert **lowercase letters to uppercase**.
8. Write a LEX program to count **spaces and tabs** in input.
9. Write a LEX program to identify **special characters**.
10. Write a LEX program to count **digits, alphabets, and special symbols** separately.
11. Write a LEX program to check whether a string is a **valid identifier**.
12. Write a LEX program to recognize **keywords**.
13. Write a LEX program to count **operators and operands**.
14. Write a LEX program to detect **palindrome strings**.
15. Write a LEX program to recognize **integer and floating-point numbers**.
16. Write a LEX program to recognize whether input is **even or odd number**.

Note:

- **VIVA will be based on Complete Syllabus**
- **Every student will get 2 programs (1 LEX program and 1 Java/Python/C/C++ program)**
- **Marks of Practical Exam will be based on VIVA, successful execution of code and proper explanation of logic of the program**